SALES PERFORMANCE ANALYSIS USING EXCEL (POWER QUERY, POWER PIVOT & DAX)

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1. INTRODUCTION

In today's data-driven world, organizations rely on data analytics to make informed decisions. This project focuses on analyzing sales performance using Microsoft Excel by leveraging advanced features such as Power Query, Power Pivot, and DAX.

The objective is to transform raw data into meaningful insights through data cleaning, modeling, and visualization.

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2. OBJECTIVE

- To analyze overall sales performance
- To calculate key business metrics like revenue, cost, and profit
- To build an interactive dashboard for decision-making
- To apply real-world data analytics concepts using Excel

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3. DATASET DESCRIPTION

The project uses three datasets:

3.1 Sales Data

Contains transaction-level data:

- OrderID
- OrderDate

- CustomerID
- ProductID
- Quantity
- UnitPrice
- TotalSale
- Region

3.2 Products Data

Contains product details:

- ProductID
- ProductName
- Category
- Cost

3.3 Customers Data

Contains customer information:

- CustomerID
- CustomerName
- Region
- Segment

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4. DATA CLEANING (POWER QUERY)

Data preprocessing was performed using Power Query:

- Removed unnecessary rows and errors
- Promoted first row as headers
- Corrected data types (text, number, date)

- Validated revenue calculation

This ensured high-quality, structured data for analysis.

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5. DATA MODELING

A relational data model was created using Power Pivot.

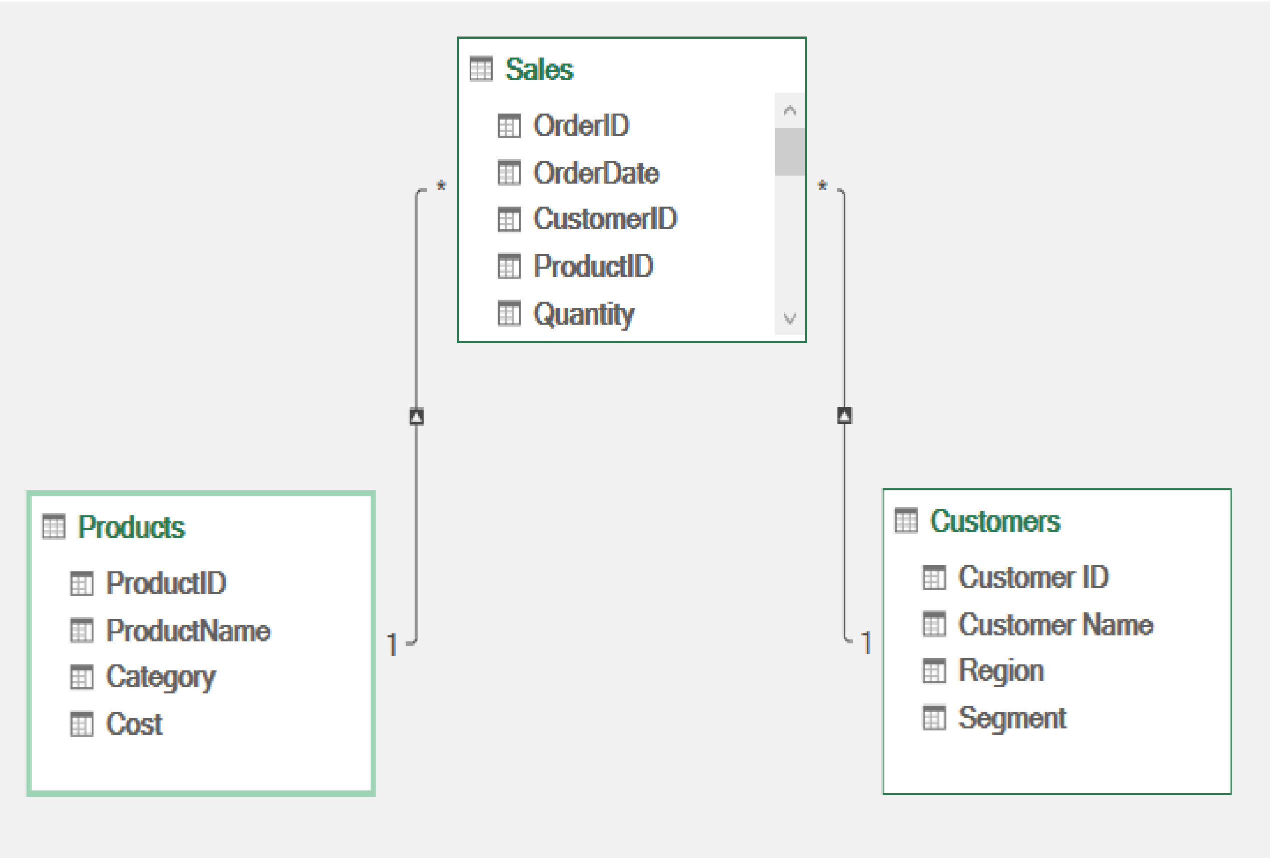
Relationships:

- Sales[ProductID] → Products[ProductID]
- Sales[CustomerID] → Customers[CustomerID]

Model Type:

Star Schema

This structure improves performance and enables advanced calculations.



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6. DAX MEASURES

Key business metrics were created using DAX:

- Total Revenue = SUM(Sales[TotalSale])
- Total Quantity = SUM(Sales[Quantity])
- Total Orders = DISTINCTCOUNT(Sales[OrderID])
- Total Cost = SUMX(Sales, Sales[Quantity] * RELATED(Products[Cost]))
- Profit = [Total Revenue] - [Total Cost]
- Profit % = DIVIDE([Profit], [Total Revenue])

These measures enable dynamic and scalable analysis.

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7. DASHBOARD DEVELOPMENT

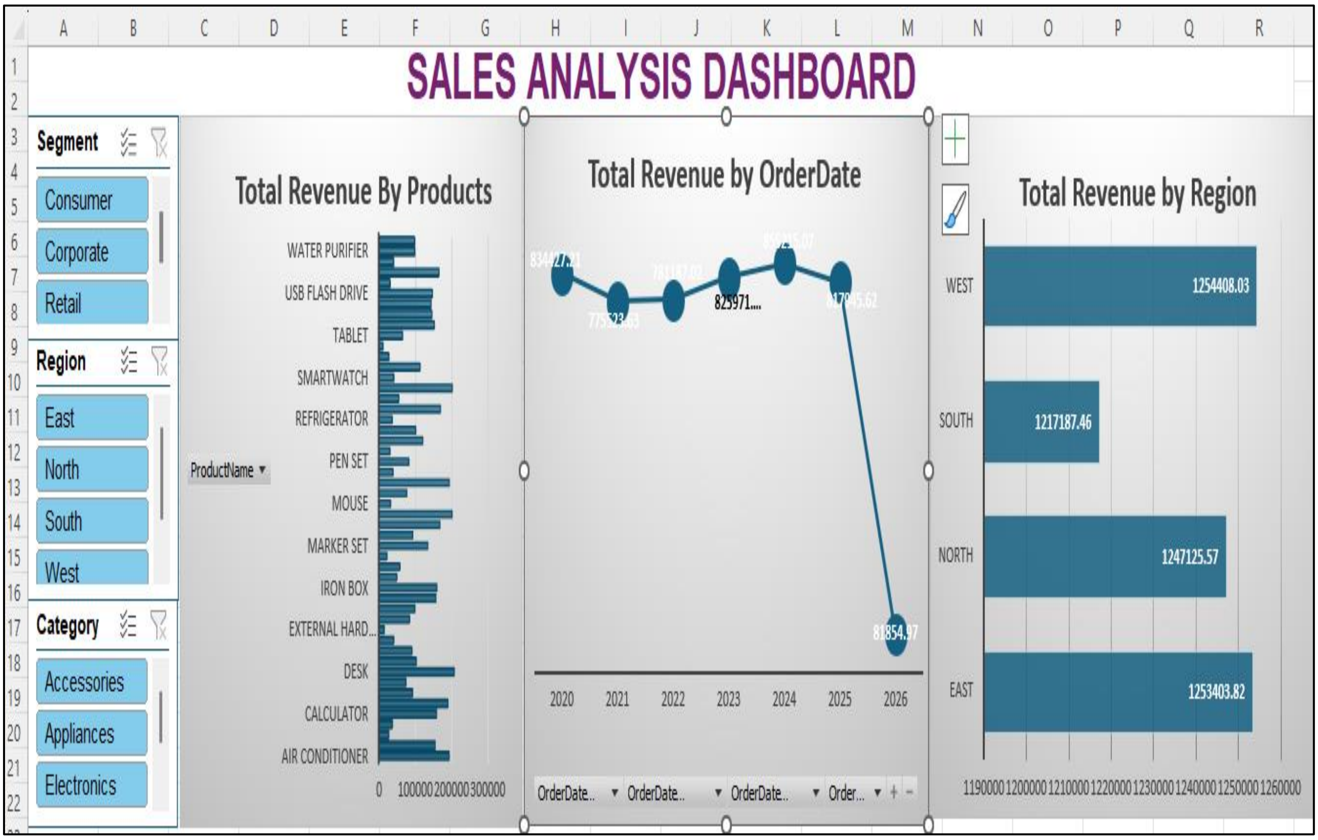
An interactive dashboard was built using Pivot Tables and Charts.

Key Components:

- KPI Cards:
  - Total Revenue
  - Profit
  - Profit %
- Charts:
  - Sales Trend (Line Chart)
  - Region-wise Sales (Bar Chart)
  - Top Products Analysis
- Slicers:
  - Region

- Category
- Segment

The dashboard allows users to filter and analyze data dynamically.



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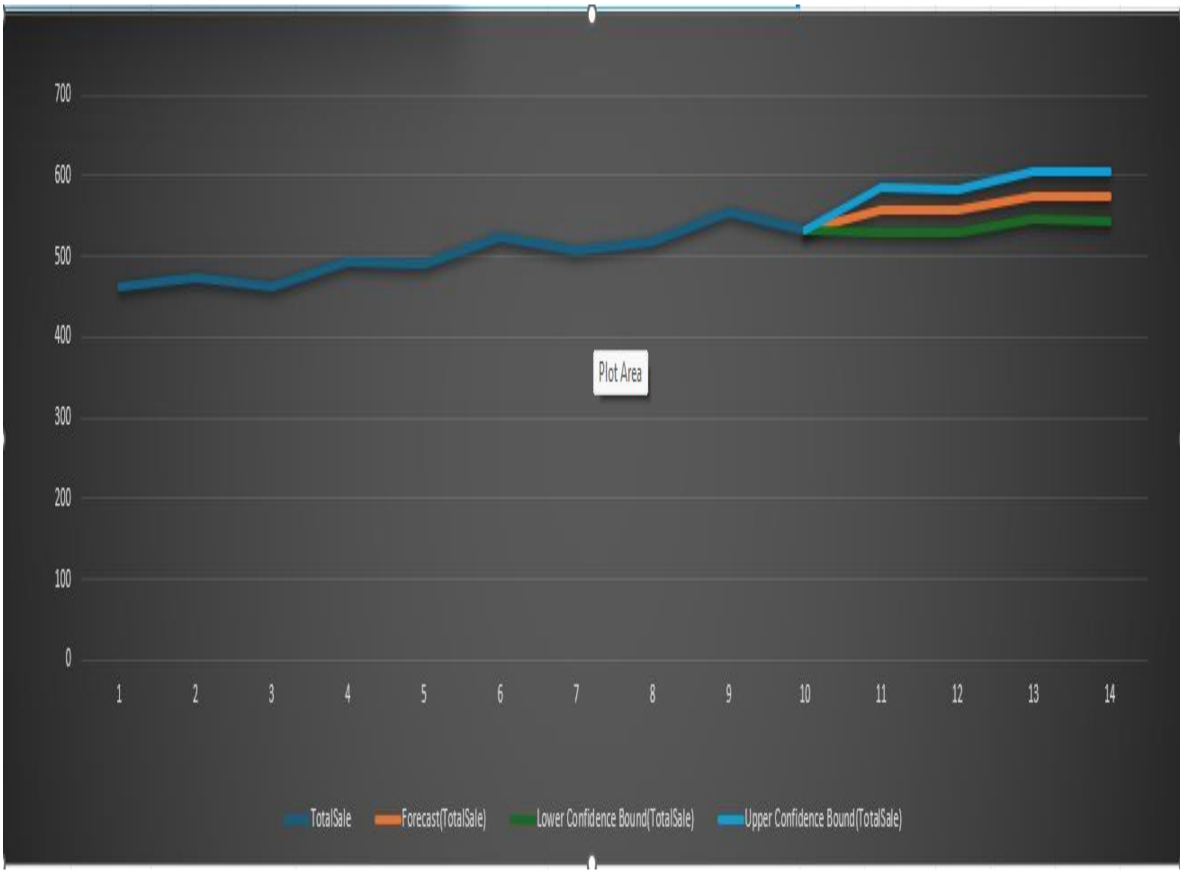
8. KEY INSIGHTS

- Identified high-performing regions
- Analyzed top-selling products
- Evaluated profitability trends
- Enabled quick decision-making using interactive filters
- **Forecast Analysis:**
- The sales data shows a consistent upward trend over time, indicating steady business growth. Using Excel's forecasting model, future sales are predicted to continue increasing.

- The forecast values range between 530 and 575, with confidence intervals indicating that actual sales are expected to fall within this range.
- The error metrics (MAE, RMSE, SMAPE) are low, suggesting that the forecasting model is reliable.
- Additionally, no significant seasonality pattern is observed, indicating stable demand without periodic fluctuations.



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9. CONCLUSION

This project demonstrates how Excel can be used as a powerful data analytics tool. By combining Power Query, Data Modeling, and DAX, a complete end-to-end analytics solution was developed.

The project highlights practical skills required for roles such as Data Analyst and Business Analyst.

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10. TOOLS USED

- Microsoft Excel
- Power Query
- Power Pivot
- DAX
- Conditional Formatting

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11. FUTURE ENHANCEMENTS

- Integration with Power BI for advanced visualization
- Adding time intelligence (Year-over-Year growth)
